When Bagging Beats Boosting: Single Trees, Bootstrap Aggregation, and SAMME on Raw Olivetti Faces

Kumar Gaurav
Roll No. P26AI0006
Machine Learning Assignment 2
Repository: REPOSITORY_URL_TO_BE_ADDED
Commit: UNCOMMITTED

Abstract

We study closed-set face identification from raw 64×64 *Olivetti images under a frozen, roll-number-seeded split of six training, two validation and two test images per identity. We compare a single CART tree, bootstrap aggregation (bagging) with up to 200 trees, and a from-scratch SAMME multiclass AdaBoost. Every choice is made on clean validation accuracy; a centred* 16×16 *block is zeroed only at test time. The selected single tree overfits (training accuracy 1.0000, 41 leaves) and reaches 0.6000 validation and 0.5000 test accuracy. Bagging with* $B = 200$ *reaches 0.9250 validation and 0.8375 clean test accuracy, and retains 0.7375 when the central block is zeroed. SAMME with depth 4 and 50 rounds reaches 0.4375 validation and 0.3875 test accuracy, and 0.2750 masked. Thirty within-identity bootstrap resamples show that bagging cuts measured variance by a factor of 30.9, while squared bias rises from 0.4208 to 0.5270; the variance reduction dominates and refit accuracy improves from 0.4687 to 0.7413. We report the unsuccessful and ambiguous outcomes as well: boosting's hard-vote multiclass decisions remain weak on 240 training faces, and in a masked case where bagging still gives the true identity 49 of 200 votes, boosting's normalised score for that identity is only 0.1484.*

1. Introduction

Face identification from raw pixels is a deliberately unfriendly test bed for classical classifiers. Each image is a 4096-dimensional vector, only 240 images are available for training, and the forty identities are separated by subtle lighting, pose and expression changes. A decision tree can fit this training set perfectly by isolating individual faces, but such a tree has no reason to generalise. Ensembling is the natural response: bagging attacks variance by averaging trees fitted to bootstrap resamples, while boosting attacks bias by reweighting examples that the current weak learners get wrong.

This report answers the three questions of Machine Learning Assignment 2 under one fixed protocol. The split is generated once from the seed obtained by reading every numeric character of the roll number `P26AI0006` in order, namely 260006, and is reused by every experiment. The split assigns six training, two validation and two test images to each of the forty identities. Model selection uses clean validation accuracy only. The clean test set is touched only for the final reporting requested by the assignment, and the masked test set is never used for training or selection. No PCA or other dimensionality reduction is applied.

Our contributions are: (i) an exact, saved split and a complete Q1–Q3 pipeline with deterministic seeds and artifact checks; (ii) student implementations of bagging and SAMME, including explicit rejection of weak learners that are no better than random; (iii) a diagnostic study of out-of-bag error, pairwise disagreement, vote margins, sample-weight concentration and masked-test behaviour; and (iv) a measured bias–variance decomposition over 30 within-identity bootstrap resamples. The headline is that bagging is the clear winner here, while SAMME is stable but weak: its weighted training error falls from 0.8500 to 0.6143, yet its hard-vote accuracy on the test set is 0.3875, below the single tree's 0.5000.

2. Protocol and Reproducibility

2.1. Data and frozen split

We use `sklearn.datasets.fetch_olivetti_faces` [9]: 40 identities, ten images each, 64×64 pixels, values in $[0, 1]$. The compact cache is stored as `data/olivetti_faces.npz`; its image array hashes

to `e6b9b0fe62f6`. The seed rule gives

$$\text{seed} = \text{all digits of } \texttt{P26AI0006} \text{ in order} = 260006. \quad (1)$$

A single `numpy.random.default_rng(260006)` stream is consumed identity by identity in ascending order. For each identity the ten dataset indices are permuted once; the first six become training images, the next two validation images and the last two test images. The resulting 240/80/80 partition has SHA-256 `8903d5baad04` and is saved to `results/split.npz` and `results/split.csv`. Every script loads this file; the split is never regenerated inside an experiment.

2.2. Masked test set and selection discipline

For the masked condition we copy the test matrix and set rows 24–39 and columns 24–39 of every image to zero, i.e. a centred 16×16 block. All other pixels are unchanged. Training and validation images remain clean. The mask is applied only when evaluating a fitted model. This is a response test, not a training augmentation.

Every hyperparameter and every ensemble size is selected by clean validation accuracy. Ties are resolved by the first row in a fixed grid order (shallower depth, then smaller leaf, then smaller B or fewer rounds). The clean test column is reported after selection; it is not used to choose a model. The masked test column is reported for the selected models and for the bagging B-grid, but never enters a selection rule.

2.3. Implementation

`scikit-learn`'s [8] `DecisionTreeClassifier` is the only base learner, as permitted by the assignment. Bagging and SAMME are implemented in `src/common.py`. The bagging implementation draws one nested sequence of 200 bootstrap replicates and 200 tree seeds; an ensemble of size B is exactly the first B trees of the 200-tree ensemble. This makes the B-curve paired rather than a collection of independently seeded runs. SAMME fits each tree with the current sample weights, records the weighted error, α_t and the weight distribution, and stops explicitly if a learner is no better than random. The full pipeline is `python -m src.run_all`; the run log, the resolved configuration, a pip freeze and SHA-256 manifests of every result file are stored under `results/`. The complete run took 1.34 hours on the run machine.

3. Q1: Single Trees and Bagging

3.1. Decision-tree grid

The grid crosses maximum depth $\{2, 4, 8, 16, \text{unrestricted}\}$ with minimum leaf size $\{1, 2, 4\}$, giving fifteen trees. Figure 1 plots validation accuracy and the selected leaf size. Accuracy rises monotonically with depth; shallow trees underfit badly (depth 2 reaches 0.0750 validation accuracy, depth 16 reaches 0.2875), and the unrestricted tree is the only one that crosses 0.5. The selected configuration is unrestricted depth with `min_samples_leaf`= 1: training 1.0000, validation 0.6000, test 0.5000. It has 41 leaves, 5.85 training samples per leaf on average, and a singleton-leaf fraction of 0.024. Training accuracy of 1.0000 with only 240 samples is the signature of memorisation, not a model of faces. The complete grid is in `results/tree_grid.csv`.

3.2. Bagging and the ensemble-size curve

Bagging [1, 2] fits B trees to bootstrap resamples of the 240 training images and predicts by plurality vote. Table 1 and Figure 2 report $B \in \{1, 5, 10, 25, 50, 100, 200\}$. A single bootstrap tree ($B = 1$) reaches 0.4625 validation and 0.3875 test accuracy, which is worse than the tree fitted to all training data; one bootstrap sample is a poor classifier. Averaging fixes this quickly: $B = 10$ gives 0.7625 validation, $B = 50$ gives 0.8500, and $B = 200$ gives 0.9250 validation and 0.8375 test accuracy. Training accuracy is already 1.0000 at $B = 10$; every additional tree reduces variance rather than training error. $B = 200$ is selected because it has the highest validation accuracy; $B = 100$ is close (0.9000 validation) and reaches the same 0.8375 test accuracy.

3.3. Out-of-bag error, disagreement and margin

Out-of-bag accuracy at $B = 200$ is 0.8042, close to the 0.8375 test estimate and well above the single tree's 0.5000. Coverage is 1.0: every training image is out of bag for at least one of the 200 trees, so the OOB estimate uses all 240 training images rather than a shrinking subset. The mean number of distinct training indices per bootstrap sample is 151.90, against the theoretical 151.89; the implementation therefore draws the intended with-replacement samples.

Trees disagree on 0.7754 of validation pairs on average. This is healthy diversity, not noise: if every tree made the same error, averaging could not help. The mean winning-minus-runner-up vote margin is 0.3023. A margin of roughly 0.30 means the winning identity holds about 30 percentage points more votes than the second identity, which is a useful confidence signal. The relationship is not monotone in B (the margin moves from 0.2675 at $B = 5$ to 0.3023 at $B = 200$) because adding trees can move both the winner and the runner-up. The three diagnostics do not tell one story: OOB accuracy and vote margin improve with B, while pairwise disagreement jumps to about 0.78 by $B = 5$ and then stays almost flat. Disagreement measures diversity, not correctness; OOB accuracy and margin are the quantities that track accuracy.

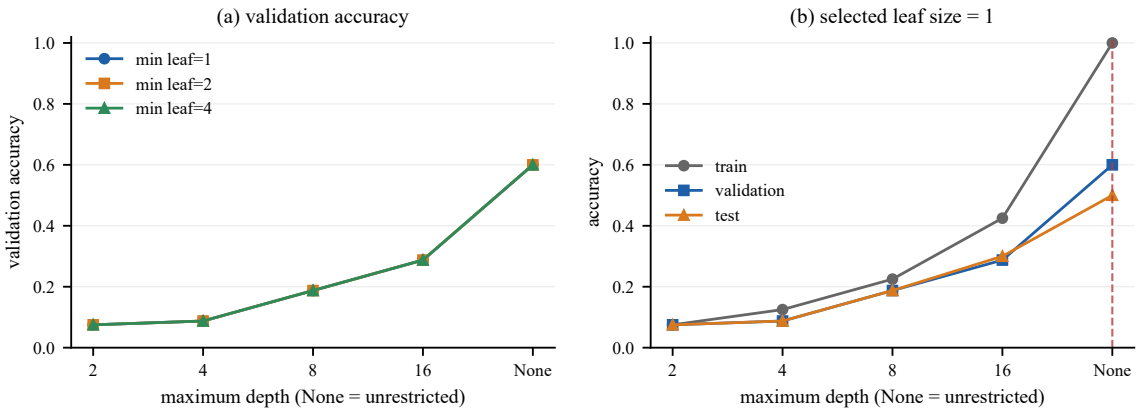


Figure 1. Q1 tree grid. (a) Validation accuracy for the fifteen depth × leaf-size combinations; unrestricted depth is the only setting above 0.5. (b) Train, validation and test accuracy at the selected leaf size 1. The dashed line marks the selected unrestricted tree. Training accuracy reaches 1.0000 while validation stalls at 0.6000.

Table 1. Bagging with the frozen tree setting. All B values use the same nested bootstrap sequence; OOB accuracy is computed only on covered training samples. Disagreement and margin are on the clean validation set. "Distinct" is the mean number of distinct training indices per bootstrap sample; the theoretical value $n(1-(1-1/n)^n)$ is in parentheses.

B	Train	Val	Test	Masked	OOB	Disagreement	Margin	Distinct (theory)
1	0.7167	0.4625	0.3875	0.0250	0.2766	0.0000	1.0000	146.0 (151.9)
5	0.9708	0.6625	0.5500	0.4125	0.3146	0.7787	0.2675	151.8 (151.9)
10	1.0000	0.7625	0.6375	0.5000	0.4008	0.7633	0.2738	153.1 (151.9)
25	1.0000	0.8250	0.7375	0.4500	0.6042	0.7741	0.2895	151.6 (151.9)
50	1.0000	0.8500	0.7875	0.6625	0.7250	0.7776	0.2887	151.5 (151.9)
100	1.0000	0.9000	0.8375	0.6625	0.7708	0.7779	0.2928	151.7 (151.9)
200	1.0000	0.9250	0.8375	0.7375	0.8042	0.7754	0.3023	151.9 (151.9)

3.4. Clean versus masked test, and face cases

The masked column of Table 1 is the most striking Q1 result. A single bootstrap tree falls to 0.0250 masked accuracy: it has learned splits on the central pixels and has no redundancy when those pixels are removed. Five trees already recover 0.4125; 200 trees reach 0.7375, only 0.1000 below their clean accuracy. Averaging many trees makes the prediction robust to the destruction of one local region even though the individual trees are not region-invariant.

The confidence numbers reinforce the difference. On clean images the selected tree's mean winning confidence is 1.0000 and bagging's is 0.3934; under the mask they are 1.0000 and 0.2657. The tree is maximally confident in both conditions, including when it is wrong, while bagging's confidence falls with its accuracy. Vote fractions are not calibrated probabilities, but they are at least sensitive to the loss of evidence.

Figure 3 shows two faces where bagging corrects the tree and two where both fail. In the first correction (dataset index 376, true identity 37) the tree predicts identity 3 with probability 1.0; bagging gives identity 37 118 of 200 votes, a margin of 0.500. In the second (dataset index 175, true identity 17) the tree predicts 22; bagging gives identity 17 114 votes, margin 0.475. The two shared errors are more interesting. For dataset index 2 (true identity 0) the tree predicts 26 and bagging predicts 15 with 56 votes, margin 0.115; for dataset index 154 (true identity 15) both predict 5, bagging with 52 votes and margin 0.070. Bagging corrects 28 of the 40 test faces that the tree gets wrong; on the remaining 12 faces both models fail. The shared errors are confident and visually plausible confusions, not tie-breaking noise. This is the boundary of what bagging can fix: it reduces variance-driven errors on faces that individual trees place on the wrong side of an unstable split, but it cannot fix systematic confusions where two identities share appearance features and all bootstrap trees agree.

4. Q2: SAMME Boosting

4.1. Algorithm and numerical safeguards

SAMME [5, 10] is multiclass AdaBoost with hard-label weak learners. With $K = 40$ classes and training weights $w_i^{(1)} = 1/n$, round t fits a tree h_t on the weighted training set, computes

$$\epsilon_t = \sum_{i=1}^{n} w_i^{(t)} \mathbf{1}[h_t(x_i) \neq y_i], \tag{2}$$

and, if $\epsilon_t < 1 - 1/K$, assigns

$$\alpha_t = \log \frac{1-\epsilon_t}{\epsilon_t} + \log(K-1). \tag{3}$$

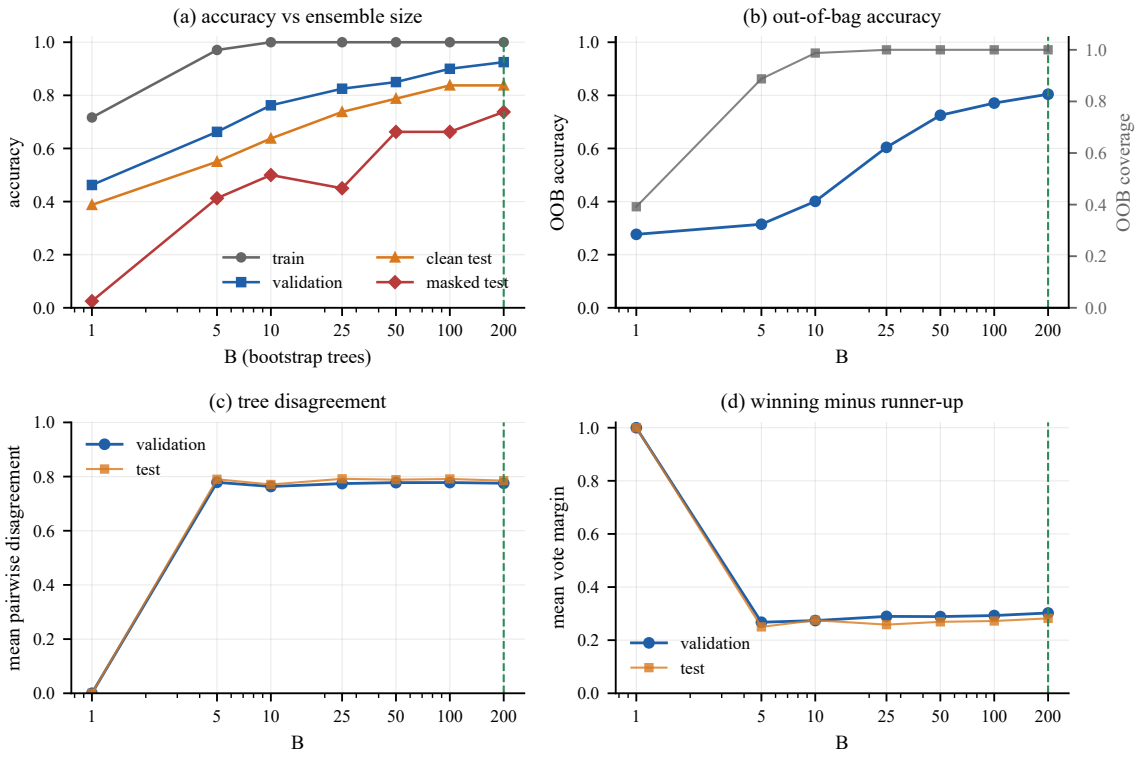


Figure 2. Bagging diagnostics. (a) Accuracy versus B on train, clean validation, clean test and masked test. (b) Out-of-bag accuracy and coverage. (c) Mean pairwise disagreement between tree predictions. (d) Mean winning minus runner-up vote margin. The dashed line marks the selected $B = 200$.

Weights are updated by

$$w_i^{(t+1)} = \frac{w_i^{(t)} \exp(\alpha_t \mathbf{1}[h_t(x_i) \neq y_i])}{\sum_{j=1}^{n} w_j^{(t)} \exp(\alpha_t \mathbf{1}[h_t(x_j) \neq y_j])}. \quad (4)$$

The ensemble score for class k is $S_k(x) = \sum_{t=1}^{T} \alpha_t \mathbf{1}[h_t(x) = k]$, and the normalised response is

$$q_k(x) = \frac{S_k(x)}{\sum_{t=1}^{T} \alpha_t}, \qquad \sum_{k=1}^{K} q_k(x) = 1. \quad (5)$$

The prediction is $\arg\max_k q_k(x)$.

Two edge cases are handled explicitly. First, if $\epsilon_t \geq 1 - 1/K$, the learner is no better than random and $\alpha_t \leq 0$; we reject that round, record the reason in `stop_reason`, and stop rather than assign a meaningless negative or zero weight. Second, if $\epsilon_t \leq \varepsilon$ with $\varepsilon = 10^{-12}$, we clip the error to ε so that α_t is finite. In this run no round was rejected: every requested round produced a learner with $\epsilon_t < 0.975$.

4.2. One complete toy round

Table 2 works through a three-class, nine-sample, one-dimensional problem with three samples per class and a depth-1 stump. The stump predicts class 1 for the three class-2 samples, so the weighted error is $\epsilon_1 = 3/9 = 1/3$ and $\alpha_1 = \log(2) + \log(2) = 1.3863$. The six correct samples are multiplied by $e^0 = 1$ and the three wrong samples by $e^{\alpha_1} = 4$; after normalisation the correct weights are $1/18 = 0.0556$ and the wrong weights are $2/9 = 0.2222$. The updated weights sum to 1.0000. This is exactly the behaviour required by Eqs. (3)–(4): the hard class-2 samples receive four times the weight of the easy samples before normalisation.

Table 2. One complete SAMME round on a three-class, nine-sample toy problem. The depth-1 stump gives weighted error $\epsilon_1 = 0.3333$ and $\alpha_1 = 1.3863$. Updated weights are normalised.

i	y_i	$w_i^{(1)}$	$h_1(x_i)$	Correct?	$w_i^{(2)}$
0	0	0.1111	0	yes	0.0556
1	0	0.1111	0	yes	0.0556
2	0	0.1111	0	yes	0.0556
3	1	0.1111	1	yes	0.0556
4	1	0.1111	1	yes	0.0556
5	1	0.1111	1	yes	0.0556
6	2	0.1111	1	no	0.2222
7	2	0.1111	1	no	0.2222
8	2	0.1111	1	no	0.2222

4.3. Depth × rounds grid

Table 3 summarises the 16-run grid (depths 1–4, requested rounds 10, 25, 50 and 100). All 16 runs accepted every

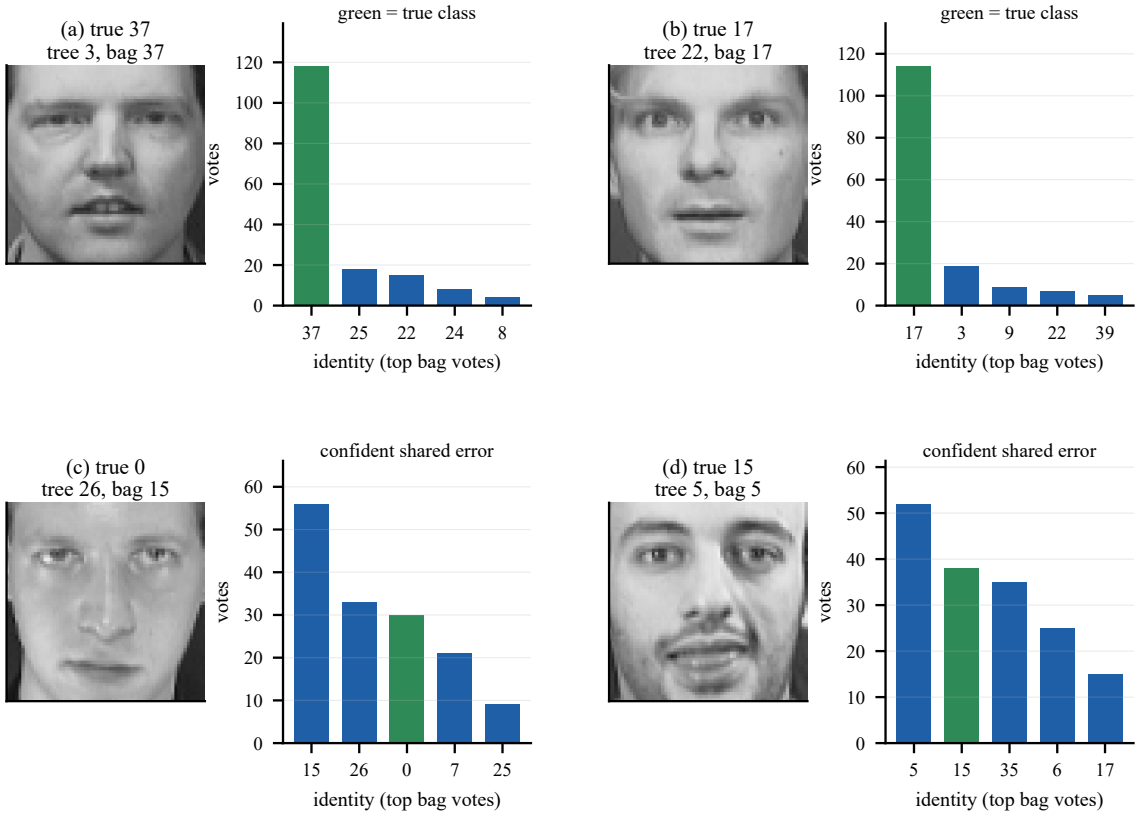


Figure 3. Q1 case studies. Top row: faces where bagging corrects the single tree (dataset indices 376 and 175). Bottom row: faces where both models fail (indices 2 and 154). Bars show the five largest bag vote counts; the true identity is green when it appears in the top five.

requested round. Depth 1 is too weak: at 100 rounds validation accuracy is only 0.1000. Depth 4 is clearly better, peaking at 50 rounds with 0.4375 validation and 0.3875 test accuracy. More rounds are not monotonically better: 100 rounds at depth 4 gives 0.4250 validation, slightly below 50 rounds, while training accuracy is 0.9708 at 50 rounds and the last weighted error is 0.6143. The selected model is therefore depth 4, 50 rounds, not the largest grid cell. Its last α_t is 3.1981, and the 100-round run's final weighted error (0.6694) is higher than the 50-round run's, so additional rounds are fitting hard faces rather than improving the ensemble.

4.4. Twenty-five, fifty and one hundred learners

Table 4 compares bagging and boosting at the three requested ensemble sizes with the frozen settings. Bagging is better at every size: 0.7375, 0.7875 and 0.8375 test accuracy at 25, 50 and 100 trees, versus 0.2875, 0.3875 and 0.3750 for SAMME. Bagging also trains faster in serial-equivalent time at 25 and 50 learners; at 100 learners the gap narrows because each boosting round refits a depth-4 tree on reweighted data. The comparison is controlled: both families see the same 240 training images, the same clean validation set and the same test set, and the bagging column uses the same nested bootstrap sequence as Table 1. The bagging times in the table are sums of per-tree fit times, not wall-clock times distorted by parallel scheduling.

Table 3. SAMME grid. Each cell is clean validation accuracy / actual accepted rounds; the full train/test/timing grid is in `results/boosting_grid.csv`. All 16 runs accepted every requested round.

Depth	10	25	50	100
1	0.037 / 10	0.062 / 25	0.075 / 50	0.100 / 100
2	0.075 / 10	0.138 / 25	0.212 / 50	0.263 / 100
3	0.113 / 10	0.175 / 25	0.287 / 50	0.325 / 100
4	0.237 / 10	0.287 / 25	0.438 / 50	0.425 / 100

4.5. The face that receives the largest weight

Figure 4 follows dataset index 79 (identity 7), the training face with the largest final SAMME weight. Its final weight is 0.0872, against a uniform weight of 0.00417—

Table 4. Controlled comparison at 25, 50 and 100 learners with the frozen settings. Bagging times are serial-equivalent sums of per-tree fit times; boosting is sequential.

Model	Learners	Train	Val	Test	Seconds
Bagging	25	1.0000	0.8250	0.7375	36.0
Boosting	25	0.5208	0.2875	0.2875	13.1
Bagging	50	1.0000	0.8500	0.7875	71.7
Boosting	50	0.9708	0.4375	0.3875	21.2
Bagging	100	1.0000	0.9000	0.8375	146.2
Boosting	100	0.9958	0.4250	0.3750	57.9

about twenty-one times larger. It is misclassified by 44 of the 50 weak learners, yet the cumulative ensemble predicts it correctly from round 1. This apparent paradox is the point of boosting: the face is hard for individual depth-4 trees, so its weight grows, but the ensemble's accumulated votes are already correct and become more so as later learners specialise on it. Figure 4(b) shows the mechanism: a correct round reduces the weight sharply, but the next tree usually errs again, so the weight climbs back, and the cycle repeats. The top ten training faces hold 0.4481 of the total weight, so reweighting is concentrated rather than uniform.

4.6. Masked response

Figure 5 studies dataset index 170 (true identity 17), a test face that all three clean models classify correctly. The full clean and masked accuracies and mean winning confidences for the three frozen models are in Table 5; here we focus on the face where the two ensembles diverge. After masking, the single tree predicts 13, bagging predicts 17 with 49 of 200 votes and margin 0.110, and boosting predicts 3. Boosting's normalised score for the true identity is only 0.148, while its score for identity 3 is 0.1807; the margin between the top two classes is 0.032. Bagging's tree disagreement rises from 0.7613 clean to 0.888 masked, so the ensemble is genuinely uncertain, but the true class still has a 49-vote lead over the runner-up. The mask removes the central 256 pixels, which is where many identity-specific contours live; bagging survives because its trees make different errors, while boosting's weighted vote is dominated by learners that have overfitted to the clean central region. Boosting's response is also shaped by its reweighted training distribution: the ten heaviest training faces carry 0.4481 of the sample weight and the single heaviest face (index 79) carries 0.0872, so later learners specialise on a handful of clean central features. When that region is removed, the α-weighted vote has no fallback, whereas bagging's unweighted plurality still has 49 trees on the true identity.

5. Q3: Measured Bias and Variance

5.1. Estimator and resampling design

For a fixed test image x_i with one-hot target e_i, let $p_r(x_i)$ be the probability vector produced by refit r and let $\bar{p}(x_i) = \frac{1}{R}\sum_{r=1}^{R} p_r(x_i)$. We measure

Table 5. Final selected models. Confidence is the mean winning vote/probability; bias^2 and variance are measured over 30 refits. The masked column uses a test set whose central 16×16 block is zeroed; training used clean images only.

Model	Clean acc.	Masked acc.	Clean conf.	Masked conf.	Bias^2	Var.
Single tree	0.5000	0.3750	1.0000	1.0000	0.4208	0.6417
Bagging	0.8375	0.7375	0.3934	0.2657	0.5270	0.0208
Boosting	0.3875	0.2750	0.1301	0.1372	0.8855	0.0206

$$\text{Bias}^2 = \frac{1}{N}\sum_{i=1}^{N}\|\bar{p}(x_i) - e_i\|_2^2, \tag{6}$$

$$\text{Var} = \frac{1}{NR}\sum_{i=1}^{N}\sum_{r=1}^{R}\|p_r(x_i) - \bar{p}(x_i)\|_2^2. \tag{7}$$

This is the assignment's Euclidean probability decomposition [6], not the zero-one loss decomposition [4, 7]; the two need not agree, and accuracy is reported separately.

We draw $R = 30$ within-identity bootstrap training sets. For each identity we sample six of its six training images with replacement, so every resample has 240 images and the same 6/2/2 identity structure as the original split. The selected tree, the 200-tree bag and the selected SAMME model (depth 4, 50 rounds) are refitted from scratch on each resample and evaluated on the same 80 clean test images. The resample indices are saved in `results/q3_resamples.npz`, and the per-refit probability arrays are saved in `results/q3_predictions.npz`.

5.2. Selected models

Table 5 and Figure 6(a) report the measured decomposition. The single tree has bias^2 0.4208 and variance 0.6417. Bagging has bias^2 0.5270 and variance 0.0208: variance falls by a factor of 30.9, while squared bias rises by 0.1063. The variance reduction dominates, so accuracy improves. This is the expected trade-off: averaging vote fractions shrinks the spread across resamples but also pulls the mean prediction away from a one-hot target. Boosting has bias^2 0.8855 and variance 0.0206. Its variance is also low, but its bias is 2.1 times the tree's, which is consistent with its low accuracy: the depth-4 weak learners are stable but not expressive enough, and the hard-vote SAMME combination does not recover the missing signal. Averaged over the 30 refits, the tree's accuracy is 0.4687 $\pm$ 0.0490, bagging's is 0.7413 $\pm$ 0.0369, and boosting's is 0.2642 $\pm$ 0.0680. The ranking is the same as on the single split; the absolute bagged mean is lower than the single-split test estimate (0.7413 versus 0.8375), which is a reminder that 80 test images and one split leave substantial uncertainty.

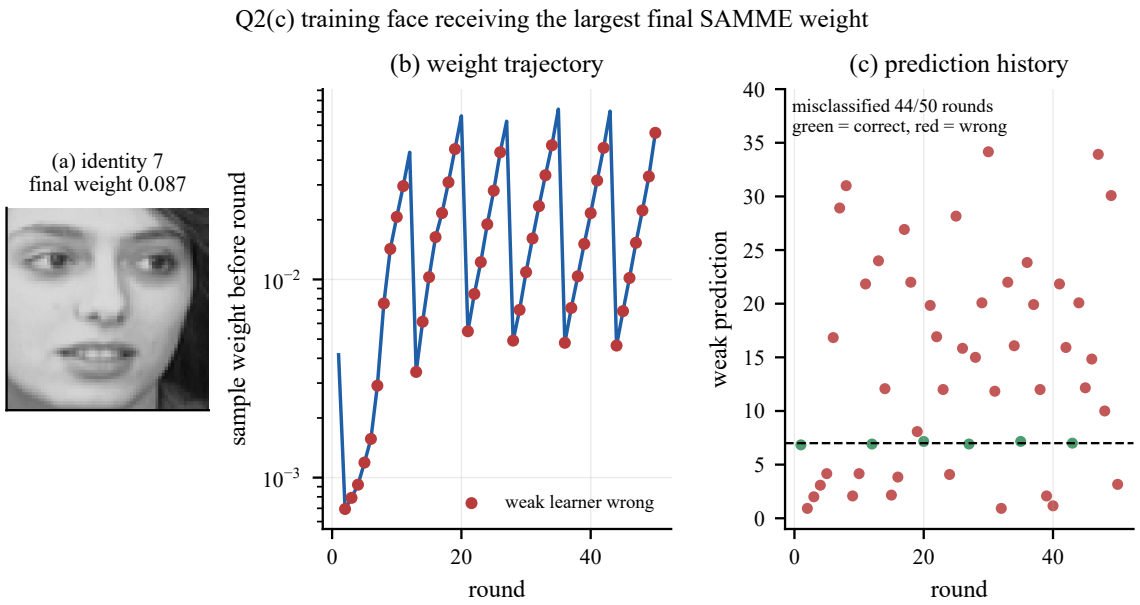


Figure 4. The training face with the largest final SAMME weight (dataset index 79, identity 7). (a) The face and its final weight. (b) Weight before each round on a log scale; red points are rounds where the weak learner is wrong. (c) Weak-learner prediction by round, with the true identity as a dashed line.

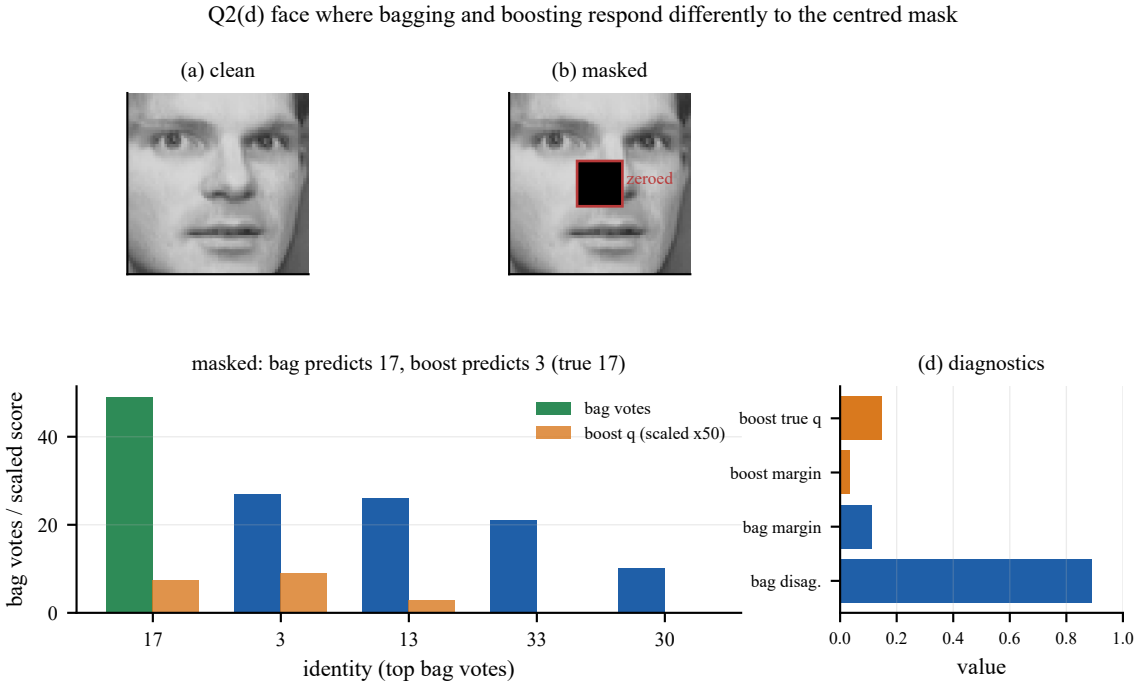


Figure 5. Masked-response case (dataset index 170, true identity 17). (a) Clean image. (b) Image with the centred 16×16 block zeroed. (c) Bag votes and scaled boosting scores for the top five identities. (d) Disagreement and margin diagnostics. Bagging still predicts 17; boosting predicts 3.

5.3. Depth study

The selected tree is unrestricted, so "deeper than selected" is not defined. We therefore compare a shallow tree (depth 2), a deep finite tree (depth 16) and the selected unrestricted tree (depth `None`), each with a matched 100-tree bag. Figure 6(c,d) gives the results.

Depth 2 has only 3.0 leaves (mean occupancy 80.0, singleton fraction 0.000); it underfits, with bias^2 0.9057, vari-

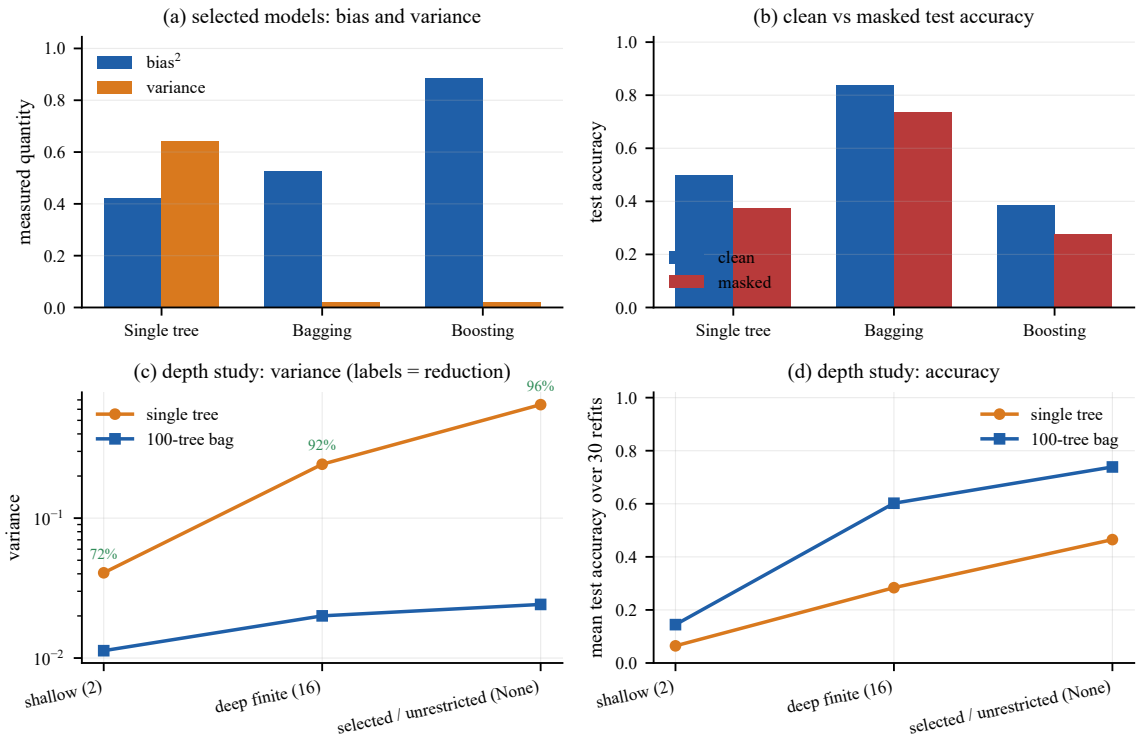


Figure 6. Q3 results. (a) Measured squared bias and variance for the selected models over 30 within-identity resamples. (b) Clean and masked test accuracy. (c) Variance of the single tree and a 100-tree bag at three depths, with the percentage reduction printed above each pair. (d) Mean test accuracy over the same 30 refits.

ance 0.0407 and accuracy 0.0646. Its bag raises accuracy to 0.1446, but the bias stays at 0.9304. Depth 16 has 18.6 leaves and variance 0.2428; the bag cuts variance by 91.8% and raises accuracy from 0.2838 to 0.6021. The unrestricted tree has 40.0 leaves (mean occupancy 6.0), variance 0.6476 and disagreement 0.823; its bag cuts variance by 96.3%, but raises bias from 0.4224 to 0.5262, and raises accuracy to 0.7387.

The largest accuracy gain is at depth 16(0.318). Thus the answer to the assignment's question is **No**: the deepest tree is not necessarily the depth at which bagging gives the largest accuracy improvement. The deepest tree is where bagging removes the most variance—96.3% versus 91.8% at depth 16 and 72.2% at depth 2—because it has the most leaves and the largest across-resample variance. But the extra variance removal is partly offset by a larger bias increase: bagging raises bias from 0.4224 to 0.5262 at the unrestricted depth, compared with 0.6394 to 0.6778 at depth 16.

The disagreement numbers rule out a simple "more diversity is better" explanation. Pairwise disagreement between the 100 bagged trees is highest for the shallow trees (0.961), lower at depth 16 (0.885) and lowest at the unrestricted depth (0.823). Deep trees all converge on the same clean-training partition, so they disagree less with each other, even though their predictions move most when the training set is resampled. Disagreement measures diversity within one ensemble; variance measures sensitivity across resamples. Leaf occupancy tells the other half of the story: depth 2 has only 3.0 leaves and cannot fit the identities, while the unrestricted tree has 40.0 leaves with mean occupancy 6.0 and the lowest singleton fraction. The depth-16 tree has 18.6 leaves, enough to fit but less extreme than the unrestricted tree, which is why its bias penalty from bagging is smaller and its accuracy gain is largest. Variance reduction is necessary for the gain, but it is not sufficient when the bias penalty also grows.

Table 5 answers the stability question directly. Boosting has the lowest variance in probability space (0.0206, marginally below bagging's 0.0208), but its refit accuracy is 0.2642 $\pm$ 0.0680, so low variance does not imply good predictions: its bias dominates. Bagging has the smallest accuracy standard deviation (0.0369) and the best clean and masked accuracy. The single tree has the largest absolute clean-to-masked drop (0.1250, versus 0.1000 for bagging and 0.1125 for boosting), and it is maximally confident in both conditions. The lowest-variance model is therefore not the best model; the best model is the one that balances the two terms, which here is bagging.

6. Discussion

Why bagging wins. The unrestricted tree has relatively low bias and high variance [3]: it can represent the training set almost perfectly, but a different bootstrap sample produces a different set of axis-aligned splits. Averaging 200 such trees removes almost all of the variance (a factor of 30.9) at the cost of a modest bias increase (0.1063), which is why bagging gains 0.3375 test accuracy over the single tree and loses only 0.1000 under the mask. The gain is not free: the bag is 200 times more expensive to fit, and its confidence is lower (mean winning vote 0.3934 clean, 0.2657 masked) because vote fractions are not calibrated probabilities.

Why boosting struggles. SAMME is designed for weak learners that are slightly better than random. With 40 classes the random threshold is 0.975, and a depth-4 tree on 240 faces has weighted error around 0.61, so it is accepted but weak. Reweighting then concentrates on a small set of hard faces (the top ten hold 0.4481 of the weight), which reduces the effective sample size and makes the next tree even more specialised. The result is a stable but biased ensemble: variance is low, bias is high, and 100 rounds do not beat 50. Boosting would likely benefit from soft class probabilities, more data, or weak learners with more capacity; those are outside the assignment's frozen protocol.

Mask behaviour. The mask is a best-case test for ensemble diversity and a worst-case test for any model that relies on the central face region. A single bootstrap tree collapses to 0.0250 masked accuracy, the selected tree drops from 0.5000 to 0.3750, bagging drops from 0.8375 to 0.7375, and boosting drops from 0.3875 to 0.2750. The ordering is unchanged, but the mechanisms differ: bagging retains a small vote lead for the true class, while boosting's normalised score flips to a different identity. The mask case is a reminder that accuracy under a fixed corruption is not a general robustness guarantee.

Limitations. The test set has only 80 images, so one image is 1.25 accuracy points; differences of a few points should not be over-interpreted. All results come from one split and one seed, as required. The 30 resamples estimate refit variability but do not turn the single split into a cross-validation study. The bias–variance numbers use the Euclidean probability decomposition and are not directly comparable to zero-one decompositions in the literature. Finally, the serial-equivalent timing is a hardware-independent proxy, not a benchmark.

7. Conclusion

Under a frozen, roll-number-seeded 6/2/2 split and raw 4096-pixel inputs, bagging is the clear winner: 200 bootstrap trees reach 0.9250 validation and 0.8375 clean test accuracy, and 0.7375 with a centred 16×16 test block zeroed. A single unrestricted tree overfits and reaches 0.5000 test accuracy. A from-scratch SAMME implementation is correct and stable—its toy round matches the closed-form weights, and its weighted error falls from 0.8500 to 0.6143—but its hard-vote multiclass decisions remain weak at 0.3875 test accuracy. The measured decomposition explains the ranking: bagging cuts variance relative to a matched tree, especially at unrestricted depth, while boosting's low variance is offset by high bias. The repository contains the saved split, all predictions and histories, the figures, the full run log, and a one-command reproduction path.

References

[1] Leo Breiman. Bagging predictors. *Machine Learning*, 24(2): 123–140, 1996.

[2] Leo Breiman. Random forests. *Machine Learning*, 45(1): 5–32, 2001.

[3] Thomas G. Dietterich. An experimental comparison of three methods for constructing ensembles of decision trees: Bagging, boosting, and randomization. *Machine Learning*, 40 (2):139–157, 2000.

[4] Pedro Domingos. A unified bias-variance decomposition and its applications. In *Proceedings of the Seventeenth International Conference on Machine Learning*, pages 231–238, 2000.

[5] Yoav Freund and Robert E. Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1): 119–139, 1997.

[6] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition, 2009.

[7] Ron Kohavi and David H. Wolpert. Bias plus variance decomposition for zero-one loss functions. In *Proceedings of the Thirteenth International Conference on Machine Learning*, pages 275–283, 1996.

[8] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[9] F. S. Samaria and A. C. Harter. Parameterisation of a stochastic model for human face identification. In *Proceedings of the Second IEEE Workshop on Applications of Computer Vision*, pages 138–142, 1994.

[10] Ji Zhu, Hui Zou, Saharon Rosset, and Trevor Hastie. Multi-

class AdaBoost. *Statistics and Its Interface*, 2(3):349–360, 2009.